TAG-NET PRESENTATION VISUAL ASSETS CATALOG

High-Resolution Images for Slides • Exact Slide Mapping & Captions

[✓] IMAGE 1: Conventional Baseline Failure (Blade Grasp)

→ Target: SLIDE 2: The Critical Failure of Conventional Grasping

→ Details: Shows conventional model (GG-CNN/Dex-Net philosophy) clamping the dangerous sharp cutting blade.

[✓] IMAGE 2: TAG-Net Architecture & AGAM Dataflow

→ Target: SLIDE 4: Proposed TAG-Net & AGAM Architecture

→ Details: Full neural dataflow: 4-channel RGB-D input, ResNet backbone, AGAM attention branch, Hadamard fusion, and 6-DoF twin.

[✓] IMAGE 3: TAG-Net Safe Knife Handover

→ Target: SLIDE 6: Safe Tool Handover Demonstration (Chef Knife)

→ Details: MuJoCo robotic arm securely grasping the knife handle and presenting it directly to the human palm.

[✓] IMAGE 4: Rotated Mug Handle Grasp (45° Wrist Roll)

→ Target: SLIDE 7: Handling Rotated Objects & Dynamic Wrist Alignment

→ Details: Robot wrist roll rotates 45° to match rotated mug, avoiding the sanitary drinking rim and grasping the loop.

[✓] IMAGE 5: Screwdriver Power Grip for Work/Torque

→ Target: SLIDE 8: Functional Tool Manipulation (Screwdriver)

→ Details: Grasping the textured rubber power grip instead of the metal shaft to apply clockwise fastening torque.

[✓] IMAGE 6: Side-by-Side Visual Proof Matrix

→ Target: SLIDE 9: Qualitative Comparison Matrix (Knife, Mug, Tool)

→ Details: Multi-tool comparison: RGB Camera vs GT Affordance Mask vs Conventional Failure (Red) vs TAG-Net Safe (Green).

[✓] IMAGE 7: Executive Web Studio Dashboard Interface

→ Target: SLIDE 10: Real-Time Web Studio & API Architecture

→ Details: Live demonstration UI showing synchronized neural heatmap perception and 6-DoF MuJoCo simulation stream.

IMAGE 1: CONVENTIONAL BASELINE FAILURE (BLADE GRASP)

WHERE TO USE IN PPT: SLIDE 2: The Critical Failure of Conventional Grasping

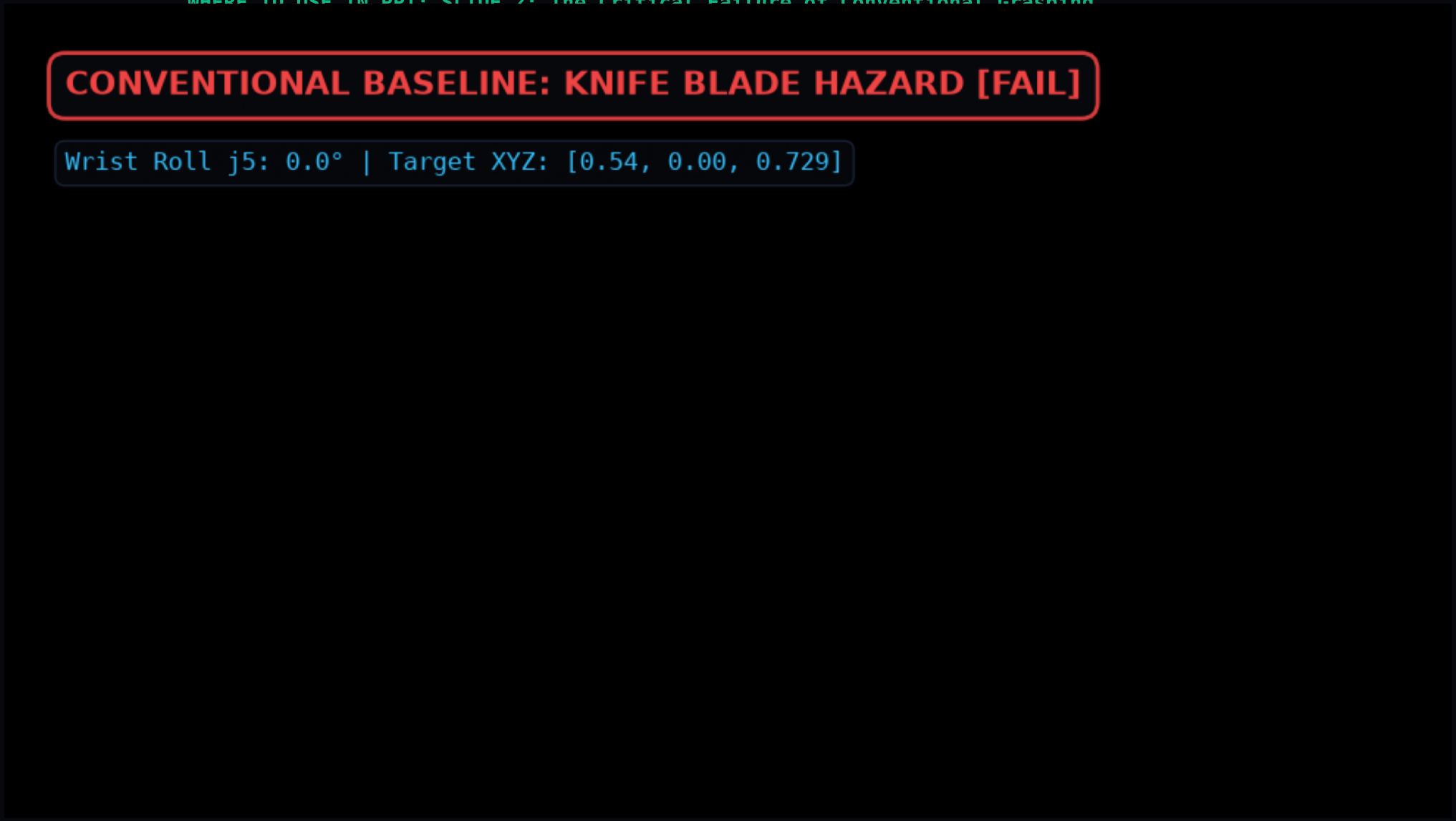


IMAGE 2: TAG-NET ARCHITECTURE & AGAM DATAFLOW

WHERE TO USE IN PPT: SLIDE 4: Proposed TAG-Net & AGAM Architecture

EXPLANATION: Full neural dataflow: 4-channel RGB-D input, ResNet backbone, AGAM attention branch, Hadamard fusion, and 6-DoF twin.

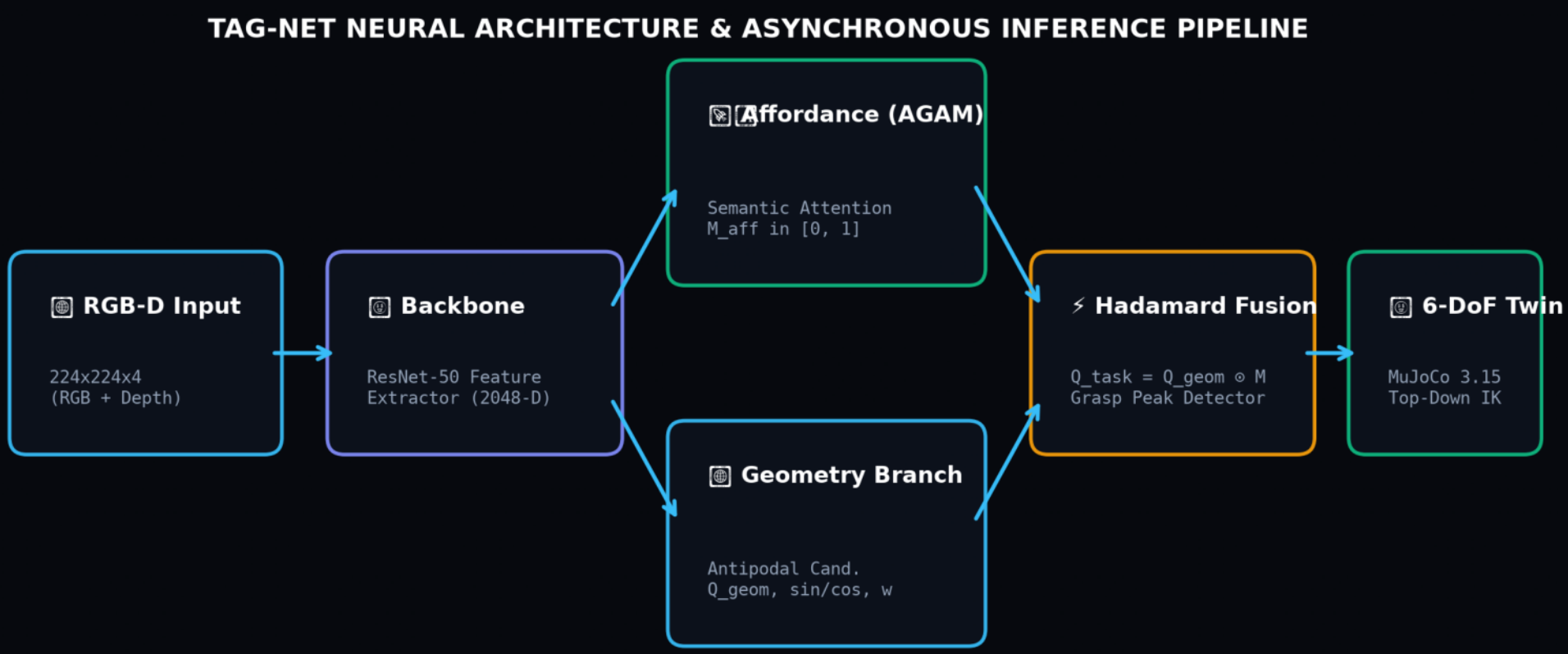


IMAGE 3: TAG-NET SAFE KNIFE HANDOVER

WHERE TO USE IN PPT: SLIDE 6: Safe Tool Handover Demonstration (Chef Knife)



IMAGE 4: ROTATED MUG HANDLE GRASP (45° WRIST ROLL)

WHERE TO USE IN PPT: SLIDE 7: Handling Rotated Objects & Dynamic Wrist Alignment

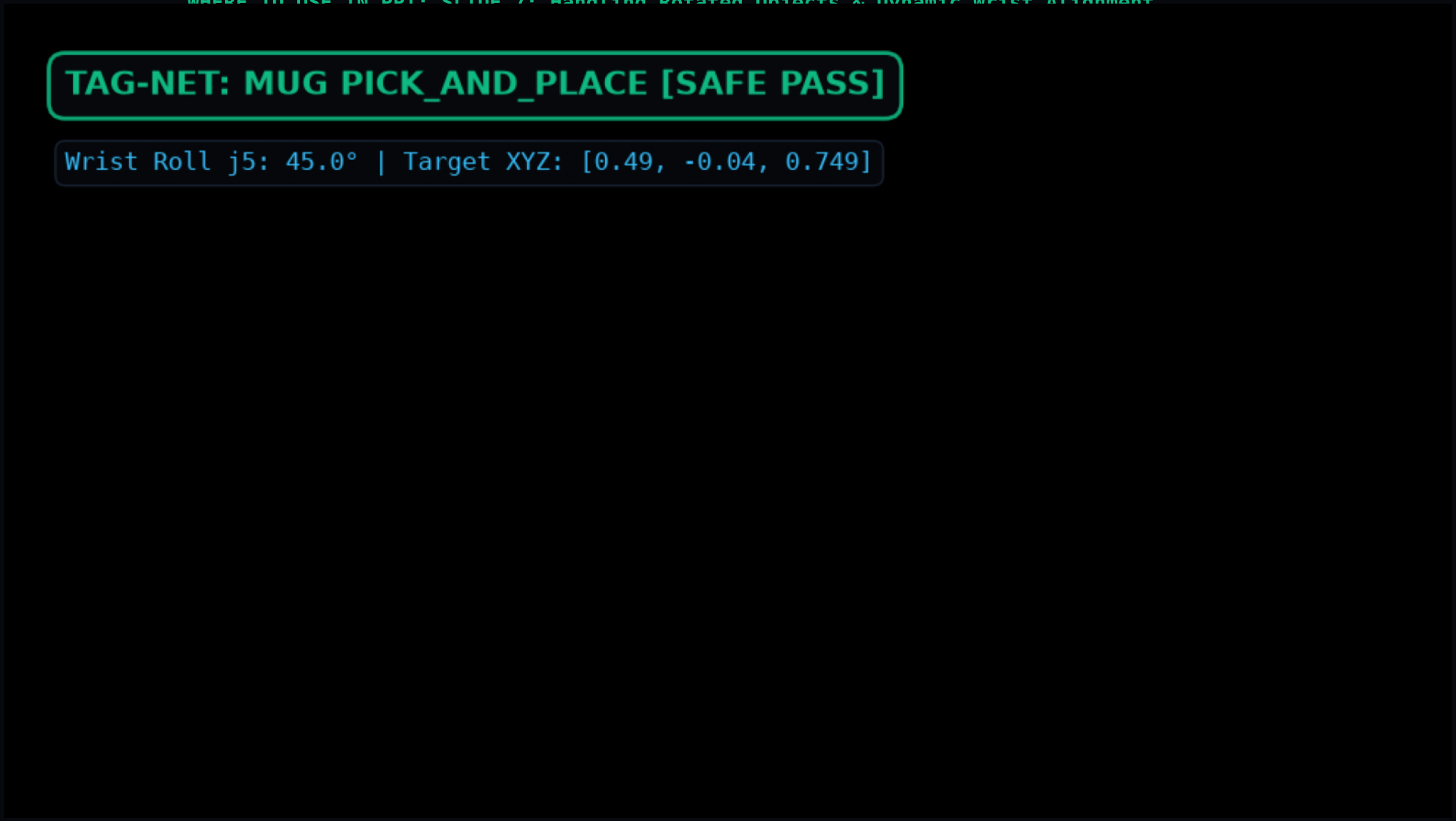


IMAGE 5: SCREWDRIVER POWER GRIP FOR WORK/TORQUE

WHERE TO USE IN PPT: SLIDE 8: Functional Tool Manipulation (Screwdriver)

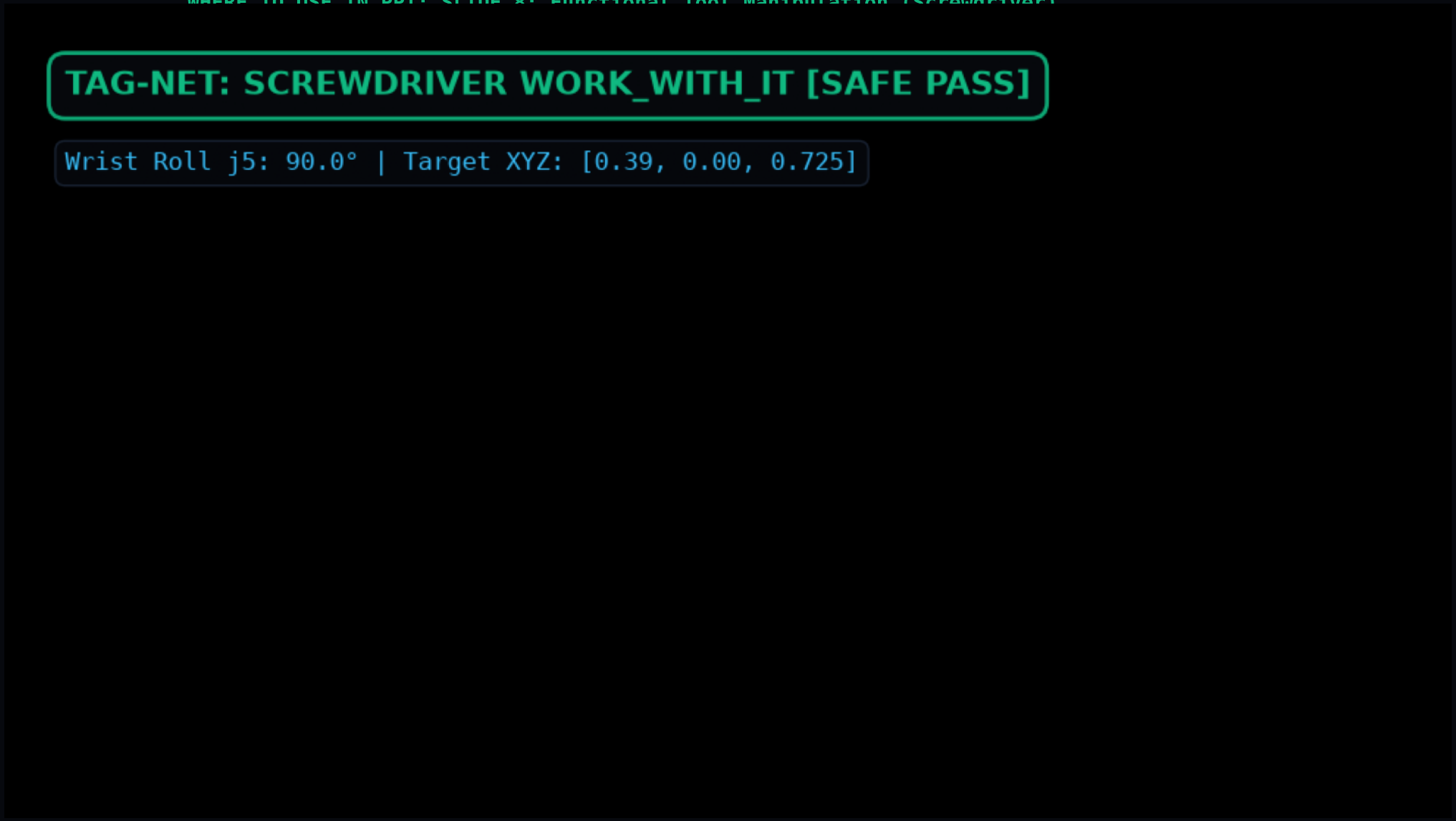


IMAGE 6: SIDE-BY-SIDE VISUAL PROOF MATRIX



IMAGE 7: EXECUTIVE WEB STUDIO DASHBOARD INTERFACE

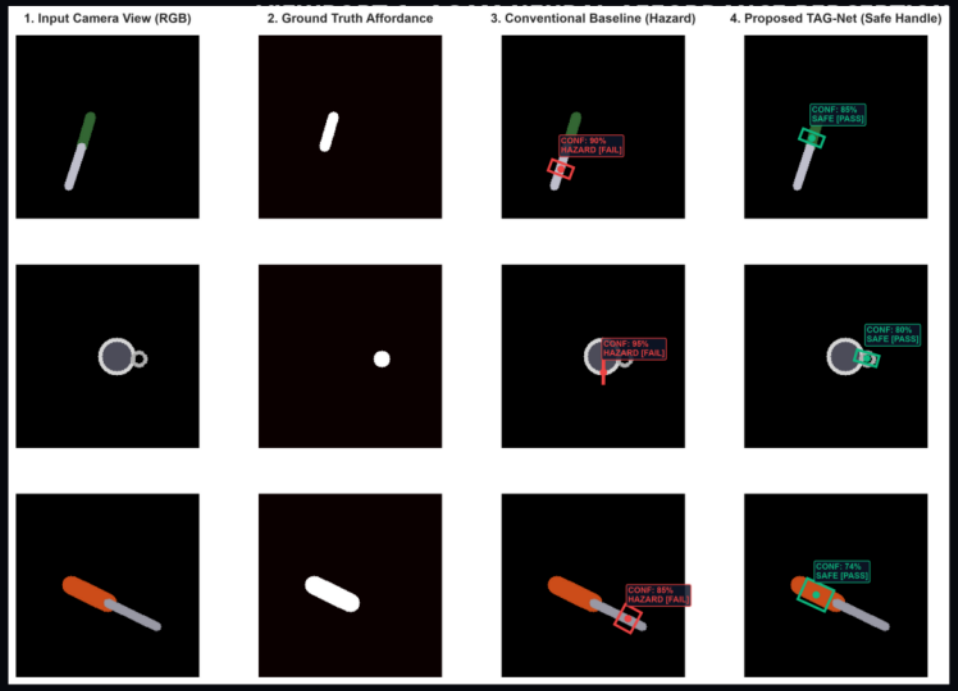
WHERE TO USE IN PPT: SLIDE 10: Real-Time Web Studio & API Architecture

EXPLANATION: Live demonstration UI showing synchronized neural heatmap perception and 6-DoF MuJoCo simulation stream.

⚡ TAG-NET STUDIO // 6-DoF ROBOTIC MANIPULATION DIGITAL TWIN

STATUS: ONLINE (56.5 | ENGINE: MUJOCO HD

FASTAPI ASYNC BACKEND • 17.69 ms INFERENCE • 56.5 FPS • MUJOCO 3.15 HD



VIEWPORT 2: 6-DoF MUJOCO PHYSICAL EXECUTION (16:9 HD)



INTERACTIVE DOCK: [MODEL: TAG-Net (AGAM)] [TOOL: Chef Knife] [TASK: Safe Handover] [ORIENTATION: 0° (Wrist Roll j5)] [α: 1.00]